

EC 101

BASICS OF ELECTRONICS & COMMUNICATION ENGINEERING

Module – V

Vol. No.-3

“Amplitude modulation”



Amplitude Modulation

□ Amplitude Modulation:

- **Amplitude Modulation (AM)** is the process of varying the amplitude of a relatively high frequency carrier signal in proportion with the instantaneous value of the information (modulating) signal, keeping the frequency and phase of the carrier constant.
- In practice, the *modulating signal* is more often an arbitrary complex analog waveform such as an audio signal.
- But a fixed-frequency sinusoidal wave (single-tone) is always considered for easy understanding and mathematical analysis purpose.
- The higher-frequency signal that is combined with modulating signal to produce the modulated waveform, is called the *carrier signal*. The high-frequency carrier signal is almost always a sinusoidal wave (unless otherwise specified) in analog modulation techniques.



Amplitude Modulation *contd.*

- Let the modulating signal be a low-frequency analog sinusoidal signal (single-tone) can be expressed as;

$$m(t) = A_m \cos(2\pi f_m t)$$

where $m(t) \rightarrow$ instantaneous value of the modulating signal (V)
 $A_m \rightarrow$ maximum(peak) amplitude of modulating signal (V)
 $f_m \rightarrow$ modulating signal frequency (Hz)

- Let the high-frequency analog carrier signal be expressed as;

$$c(t) = A_c \cos(2\pi f_c t)$$

where $A_c \rightarrow$ maximum(peak) amplitude of carrier signal (V)
 $f_c \rightarrow$ carrier signal frequency (Hz)



Amplitude Modulation *contd.*

- As per the definition of amplitude modulation, the instantaneous value of the amplitude modulated (AM) signal can be expressed as;

$$\begin{aligned}
 s_{AM}(t) &= \underline{c(t)} + \underline{k_a m(t) \cdot c(t)} \\
 &= A_c [1 + k_a m(t)] \underline{\cos(2\pi f_c t)}
 \end{aligned}
 \quad \begin{array}{l}
 [A_c + m(t)] \cos \omega_c t \\
 [A_c + \text{Am Coefficient}] \cos \omega_c t \\
 A_c [1 + \frac{A_m}{A_c} \cos \omega_m t] \cos \omega_c t
 \end{array}
 \quad \text{[Eq.5.3.1]}$$

- From the amplitude modulated signal waveform, it is observed that;
 - When no modulating signal is applied, the modulated AM signal waveform is simply the unmodulated carrier signal.
 - When a modulating signal is applied, the AM signal waveform varies in-accordance with the amplitude of the modulating signal.
 - The frequency of the carrier signal in the AM signal waveform remains the same as that of unmodulated carrier signal.



Amplitude Modulation contd.

□ The AM Envelope:

The time varying shape of the amplitude modulated waveform is called the envelope of the AM signal or simply the *AM-envelope*.

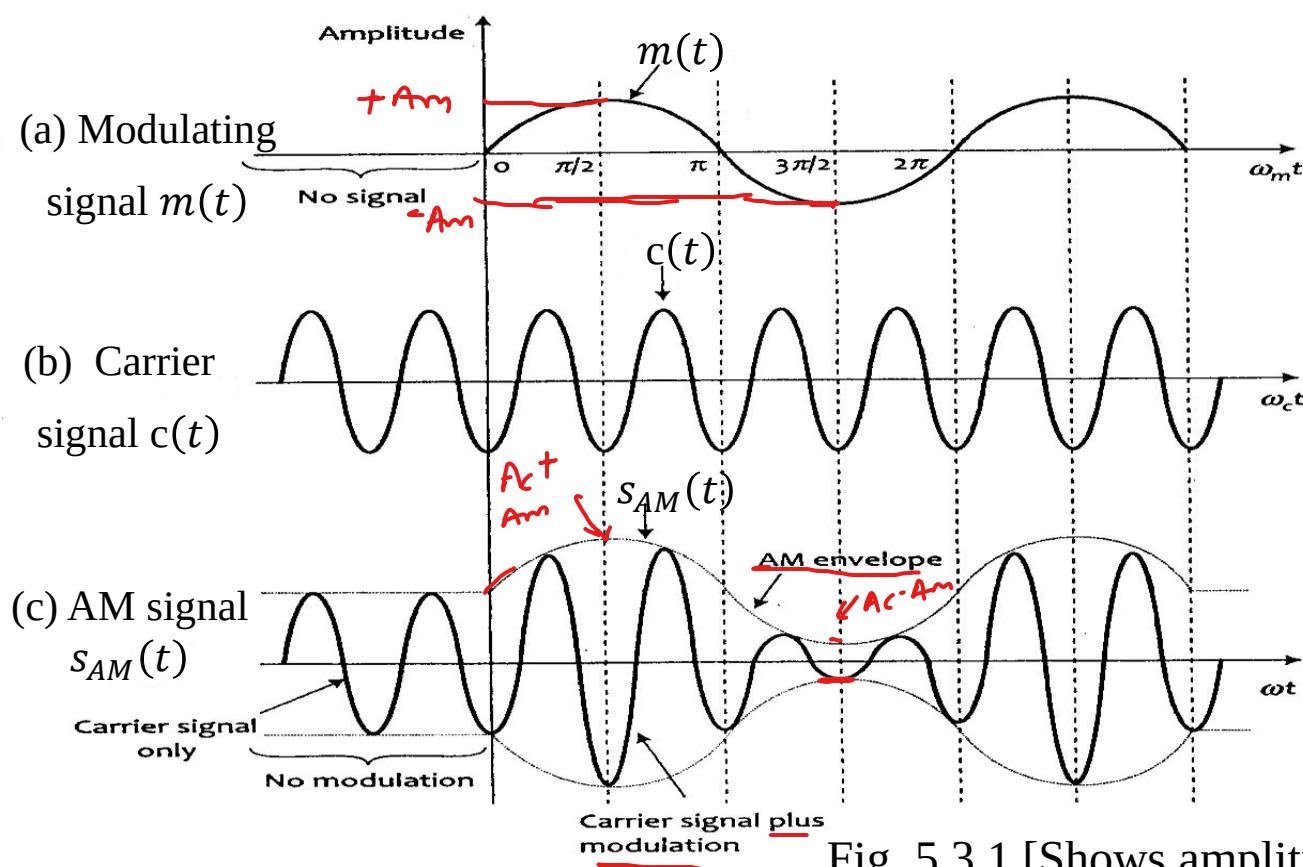


Fig. 5.3.1 [Shows amplitude modulation]

Amplitude Modulation contd.

- From the amplitude modulated waveform as shown in Fig. 5.3.1(c), the following is observed:
 - When the peaks of the individual waveforms of the carrier signal in amplitude modulated wave in positive and negative half are joined together separately (as shown by dotted lines), the resulting envelopes resemble the original modulating signal.
 - The exterior shape of the each half (positive or negative) of the AM envelope is identical to the shape of the modulating signal.
 - The modulated AM wave contains the information signal in the amplitude variations of the carrier signal.
 - The AM envelope contains all the frequency components i.e. the carrier signal, sum and difference of carrier frequency and modulating frequency.



Amplitude Modulation *contd.*

□ Time-domain Analysis of AM signal:

- An amplitude modulated (AM) wave may be described as a function of time in the form:

$$\underline{s_{AM}(t)} = A_c [1 + k_a m(t)] \cos(2\pi f_c t)$$

Where $k_a = \frac{1}{A_c} \rightarrow$ a constant called the amplitude sensitivity of the modulator

We observe, the envelope of $s_{AM}(t)$ has the same shape as the baseband signal $m(t)$ provided *two requirements* are satisfied.

- The amplitude of $k_a m(t)$ is always less than unity, i.e.

$$|k_a m(t)| < 1, \text{ for } \underline{\text{all } t}$$

It ensures that the function $[1 + k_a m(t)]$ is always positive. The envelope of AM wave $s_{AM}(t)$ may be expressed as $A_c [1 + k_a m(t)]$.



Amplitude Modulation *contd.*

- When the amplitude sensitivity k_a of the modulator is large enough to make $|k_a m(t)| > 1$ for any t , the carrier wave becomes **over-modulated**, resulting in **carrier phase reversals** whenever the factor $[1 + k_a m(t)]$ crosses zero. The modulated wave then exhibits **envelope distortion** as in Fig.5.3.2(c). It is therefore apparent that by avoiding over-modulation, a one-to-one relationship is maintained between the envelope of the AM wave and the modulating wave for all values of time. The absolute maximum value of $k_a m(t)$ multiplied by 100 is referred to as the **percentage of modulation (μ)**.
- The carrier frequency f_c is much greater than the highest frequency component f_m of the modulating/message signal $m(t)$, i.e.

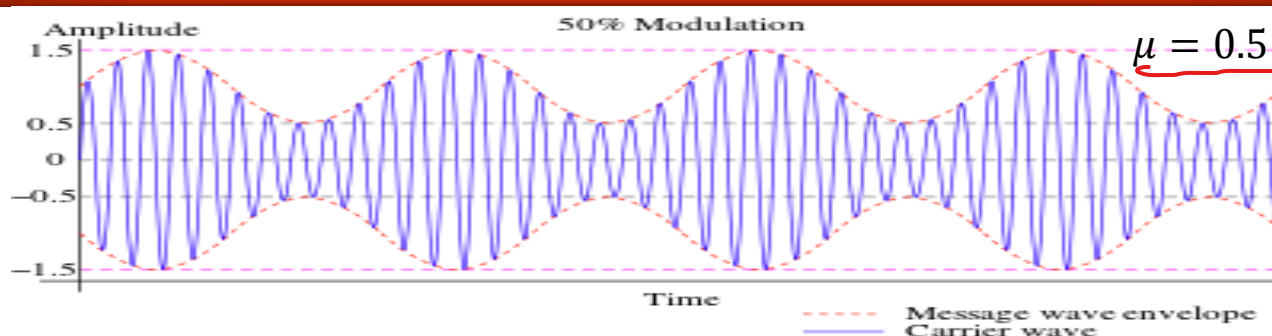
$$f_c \gg f_m \quad \checkmark$$

We call f_m the **message bandwidth**. If the above condition is not satisfied, an envelope cannot be visualized satisfactorily.

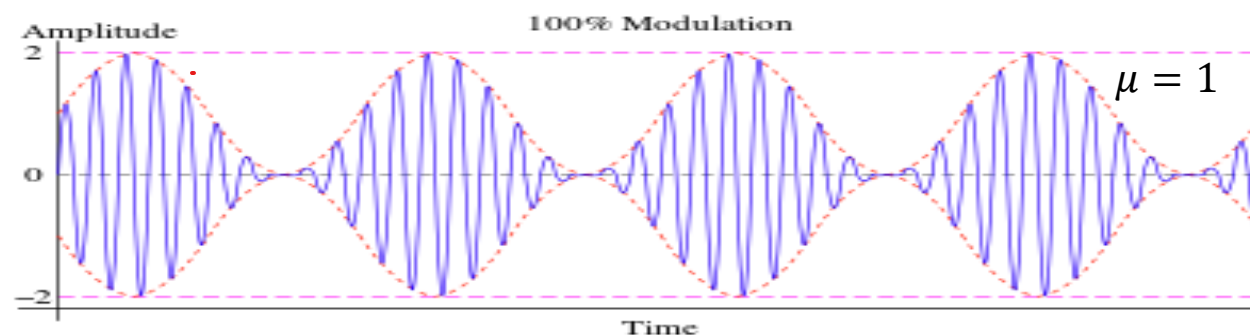


Amplitude Modulation *contd.*

Under
modulation
Fig. (a)



100%
modulation
Fig. (b)



Over
modulation
Fig. (c)

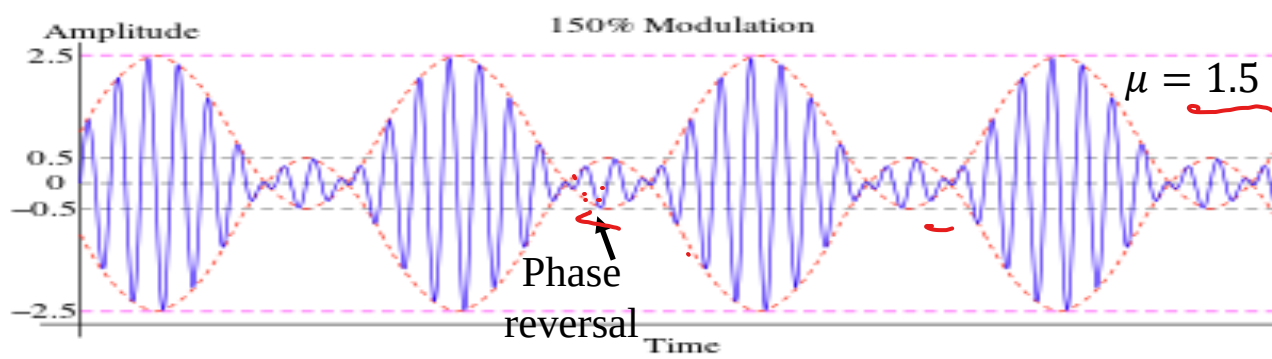


Fig.5.3.2 [Effect of modulation index on AM-envelope]

Amplitude Modulation *contd.*

□ Modulation Index:

- The ***modulation-index*** (μ) is defined as the ratio of the maximum value of the message signal amplitude to the maximum value of the un-modulated carrier signal amplitude. It is also called the ***Depth of modulation***.

- $\mu = \frac{A_m}{A_c} = \frac{m(t)|_{max}}{c(t)|_{max}} \leq 1$

- $\mu = \frac{\text{r.m.s.value of } m(t)}{\text{r.m.s.value of } c(t)} = \sqrt{\frac{(A_m/\sqrt{2})^2}{(A_c/\sqrt{2})^2}} = \frac{A_m}{A_c}$

- Also it can be found from the AM-envelope as;

$$\mu = \frac{s(t)|_{max} - s(t)|_{min}}{s(t)|_{max} + s(t)|_{min}} = \frac{(A_c + A_m) - (A_c - A_m)}{(A_c + A_m) + (A_c - A_m)} = \frac{A_m}{A_c} \quad [\text{Eq.5.3.2}]$$



Amplitude Modulation contd.

Proof of μ :

$$\therefore \mu = \frac{m(t)|_{max}}{A_c} = \frac{s(t)|_{max} - A_c}{A_c}$$

[Eq. A]

Also,

$$\mu = \frac{A_c - s(t)|_{min}}{A_c}$$

[Eq. B]

- Adding Eqn-A & B, we get, $\mu = \frac{s(t)|_{max} - s(t)|_{min}}{2A_c}$

[Eq. C]

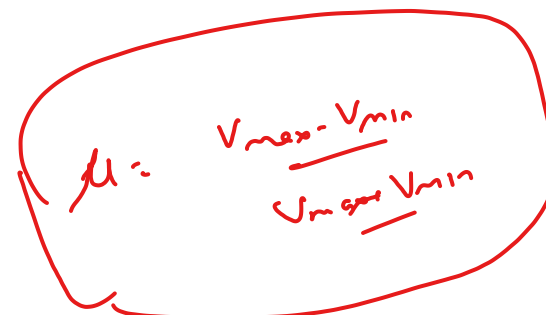
- Subtracting Eqn-B from Eqn-A, we get,

$$2A_c = s(t)|_{max} - s(t)|_{min}$$

[Eq. D]

- Substituting Eqn-D in Eqn-C, we get,

$$\mu = \frac{s(t)|_{max} - s(t)|_{min}}{s(t)|_{max} + s(t)|_{min}} = \frac{(A_c + A_m) - (A_c - A_m)}{(A_c + A_m) + (A_c - A_m)} = \frac{A_m}{A_c} \checkmark$$



$\mu = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$



Amplitude Modulation *contd.*

□ Expression of AM signal in Terms of Modulation Index:

- Re-writing the general expression of amplitude modulated (AM) signal in time domain as,

$$s_{AM}(t) = A_c [1 + k_a m(t)] \cos(2\pi f_c t) \quad \checkmark$$

For a single-tone modulation [$m(t) = A_m \cos(2\pi f_m t)$], we get,

$$\begin{aligned} s_{AM}(t) &= A_c [1 + k_a A_m \cos(2\pi f_m t)] \cos(2\pi f_c t) \\ &= A_c [1 + \underbrace{\mu}_{\mu} \underbrace{\cos(2\pi f_m t)}_{\cos(2\pi f_m t)}] \cos(2\pi f_c t) \quad \checkmark \end{aligned} \quad [\text{Eq.5.3.3}]$$

- Thus the modulation index signifies the amount by which the amplitude of the carrier signal is varied in the process of amplitude modulation.



Fourier Transform

Fourier Transform Formulae

$$\underline{G(f)} = \int_{-\infty}^{\infty} \underline{g(t) \cdot e^{-j2\pi ft} \cdot dt} = \underline{FT [g(t)]}$$

and

$$g(t) = \int_{-\infty}^{\infty} G(f) \cdot e^{j2\pi ft} \cdot df = \underline{FT^{-1}[G(f)]}$$

[Eq.5.3.4]

OR

$$\underline{G(\omega)} = \int_{-\infty}^{\infty} g(t) \cdot e^{-j\omega t} \cdot dt = FT [g(t)]$$

and

$$g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) \cdot e^{j\omega t} \cdot d\omega = FT^{-1}[G(\omega)]$$

[Eq.5.3.5]



Fourier Transform contd.



□ Existence of The Fourier Transform:

- The function $g(t)$ to be Fourier transformable, it is sufficient that $g(t)$ satisfies the ***Dirichlet's*** conditions:
- The function $g(t)$ is a single valued function with a finite number of maxima and minima and a finite number of discontinuities in any finite time interval.
- The function $g(t)$ is absolutely integrable, i.e.

$$\int_{-\infty}^{\infty} |g(t)| \cdot dt < \infty \quad [\text{Eq.5.3.6}]$$

These conditions include all energy signals for which we have,

$$\int_{-\infty}^{\infty} |g(t)|^2 \cdot dt < \infty \quad [\text{Eq.5.3.7}]$$



Fourier Transform contd.

□ Nature of Fourier Transform $G(f)$:

- In general, the Fourier transform $G(f)$ is a complex function of frequency 'f', hence may be expressed as,

$$G(f) = |G(f)|e^{j\theta(f)} \quad [\text{Eq.5.3.8}]$$

$$= |G(f)|\angle\theta(f)$$

$$= \text{Re}[G(f)] + j \text{Im}[G(f)]$$

Where $G(f)$ is the continuous magnitude spectrum of $g(t)$, and $\theta(f)$ is the continuous phase spectrum of $g(t)$.

- The **Magnitude spectrum** is drawn between $|G(f)|$ v/s (f) and is an **even function** $|G(f)| = |G(-f)|$.
- The **Phase-spectrum** is drawn between $\theta(f)$ v/s (f) and always is an **odd-function** i.e. $\theta(f) = -\theta(-f)$.



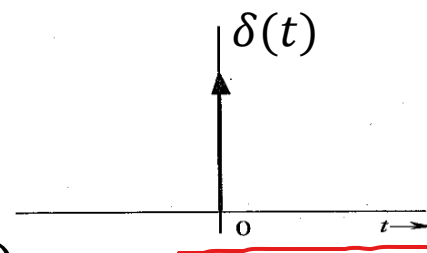
Dirac Delta Function

□ Dirac Delta Function: (Impulse)

- The Dirac delta function, denoted by $\delta(t)$, is defined as having zero amplitude everywhere except at $t = 0$ where it is infinitely large in such a way that it contains unit area under its curve; that is,

$$\delta(t) = 0, \quad t \neq 0$$

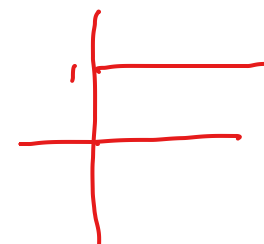
$$\text{And } \int_{-\infty}^{\infty} \delta(t) \cdot dt = 1$$



[Eq.5.3.9]

- Product of $\delta(t)$ and any time function $g(t)$,

$$\int_{-\infty}^{\infty} g(t) \cdot \delta(t) \cdot dt = g(0)$$



[Eq.5.3.10]

- The time-shifting property of delta function is,

$$\int_{-\infty}^{\infty} g(t) \cdot \delta(t - t_0) \cdot dt = g(t_0)$$

[Eq.5.3.11]



Application of Delta Function

Application: dc Signal:

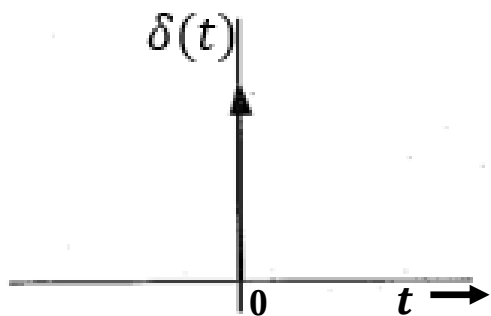
Thus from the Fig.5.3.3 we have;

$$\delta(t) \Rightarrow 1$$

And applying the Duality theorem, we obtain;

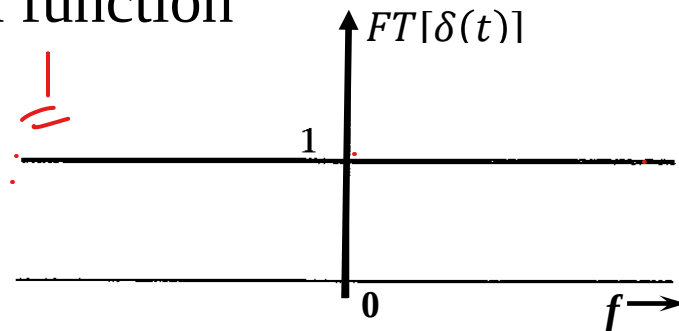
$$1 \Rightarrow \delta(-f) = \delta(f) \quad ; \text{ as } \delta(f) \text{ is an even function}$$

Thus;



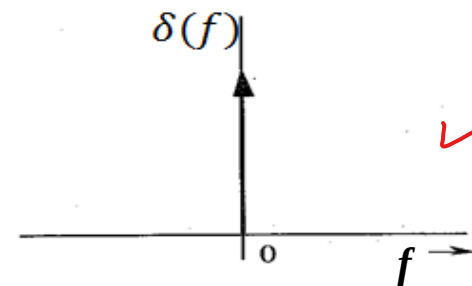
$$\left[e^{-j\omega t} \right]_{t=0} = 1$$

$\Downarrow$



$\Downarrow$

Fig.5.3.3

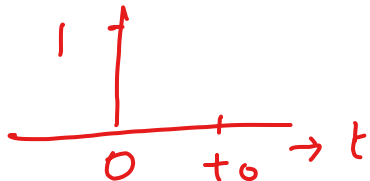


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Application of Delta Function *contd.*

Application: Complex Exponential Function :

By applying the Frequency shifting property to a constant function, we obtain another Fourier-transform pair.

i. e. $\underline{e^{j2\pi f_c t}} \cdot 1 \Rightarrow \delta(f - f_c)$  [Eq.5.3.14]

and $e^{-j2\pi f_c t} \cdot 1 \Rightarrow \delta(f + f_c)$  [Eq.5.3.15]

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{j2\pi f_c t} \cdot e^{-j\omega t} dt$$

Thus the complex exponential function $e^{j2\pi f_c t} [e^{-j2\pi f_c t}]$ is transformed in the frequency domain in a Delta function $\delta(f - f_c) [\delta(f + f_c)]$ occurring at $f = f_c$

$$F(\omega) = \int_{-\infty}^{\infty} e^{-j(\omega - \omega_c)t} dt = F(\omega - \omega_c)$$

Application of Delta Function *contd.*

■ Application: Sinusoidal Functions($\cos 2\pi f_c t$):

- Evaluation of FT of $\cos 2\pi f_c t$: By Euler's Theorem, we write:

$$\cos(2\pi f_c t) = \frac{1}{2} (e^{j2\pi f_c t} + e^{-j2\pi f_c t}) \quad [\text{Eq.5.3.16}]$$

Then, $\cos(2\pi f_c t) \Leftrightarrow \frac{1}{2} [\delta(f - f_c) + \delta(f + f_c)]$ [Eq.5.3.17]

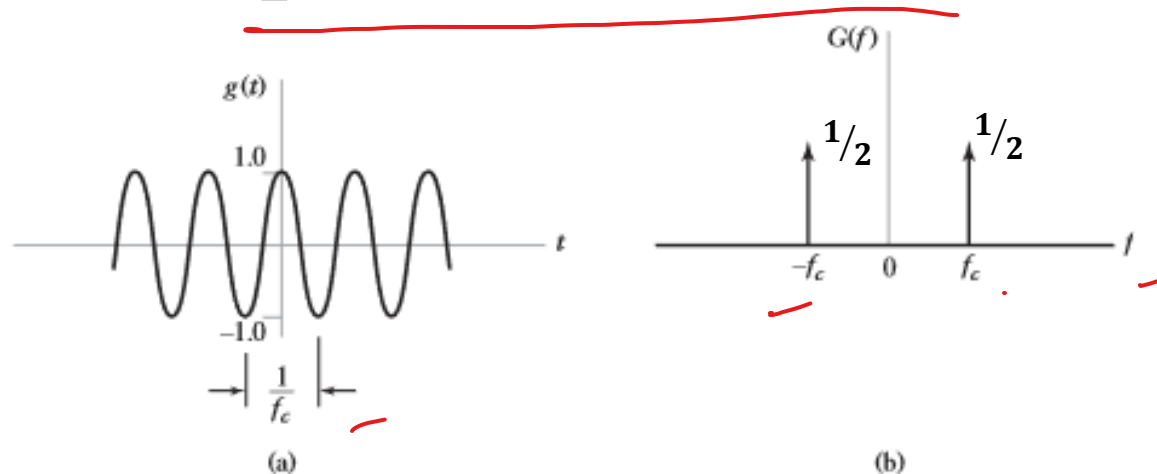


Fig.5.3.4 [(a) Cosine function, (b) Spectrum]

Application of Delta Function *contd.*

■ Application: Sinusoidal Functions($\sin 2\pi f_c t$):

- Evaluation of FT of $\sin 2\pi f_c t$: By Euler's Theorem, we write,

$$\sin(2\pi f_c t) = \frac{1}{2j} (e^{j2\pi f_c t} - e^{-j2\pi f_c t}) \quad [\text{Eq.5.3.18}]$$

$$\text{Then, } \sin(2\pi f_c t) \Leftrightarrow \frac{1}{2j} [\delta(f - f_c) - \delta(f + f_c)] \quad [\text{Eq.5.3.19}]$$

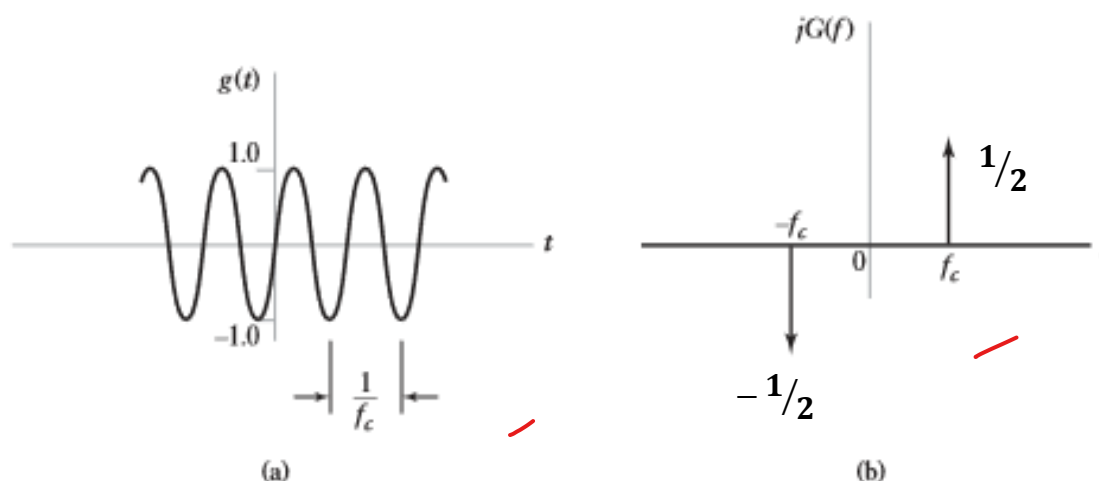


Fig.5.3.5 [(a) Cosine function, (b) Spectrum]

Frequency-domain Analysis of AM signal

□ Frequency-domain Analysis of AM signal:

- Re-writing the generic mathematical expression of an AM signal [Eq.5.3.1] as,

$$s_{AM}(t) = A_c[1 + k_a m(t)] \cos(2\pi f_c t) = \underbrace{A_c}_{\text{correct}} \cos(2\pi f_c t) + \underbrace{\frac{A_c}{2}}_{\text{correct}} [m(t) \cos(2\pi f_c t) + m(t) \sin(2\pi f_c t)]$$

- Taking the Fourier transform of the above Eqⁿ, we get,

$$S_{AM}(f) = \mathcal{F}[s_{AM}(t)] = \frac{A_c}{2} [\delta(f - f_c) + \delta(f + f_c)] + \frac{k_a A_c}{2} [M(f - f_c) + M(f + f_c)] \quad \text{[Eq.5.3.20]}$$

Where we have used the relations:

$$\cos(2\pi f_c t) = \frac{1}{2} [\exp(j2\pi f_c t) + \exp(-j2\pi f_c t)]$$

$$\exp(j2\pi f_c t) \Rightarrow \delta(f - f_c)$$

and, $m(t)\exp(j2\pi f_c t) \Rightarrow M(f - f_c); m(t)\exp(-j2\pi f_c t) \Rightarrow M(f + f_c)$

Frequency-domain Analysis of AM signal contd.

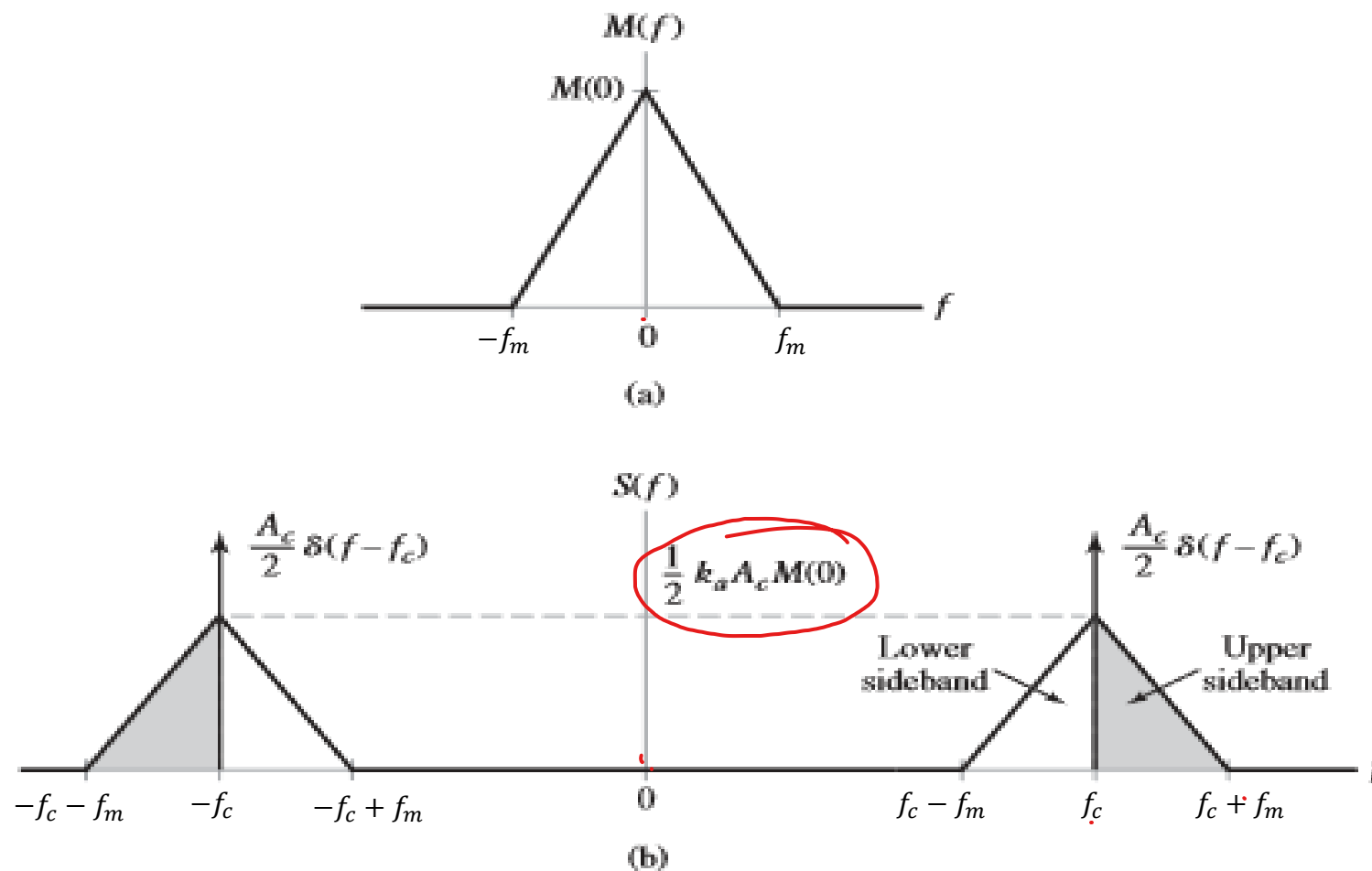


Fig.5.3.6 (a) spectrum of message signal $m(t)$, (b) spectrum of AM wave $s_{AM}(t)$

Frequency-domain Analysis of AM signal contd.

- The spectrum representation of AM signal consists of three components.
 - The un-modulated carrier components at a frequency $\pm f_c$.
 - The **upper sideband (USB)** between the frequency range of f_c to $(f_c + f_m)$ and $-f_c$ to $(-f_c - f_m)$.
 - The **lower sideband (LSB)** between the frequency range of $(f_c - f_m)$ to f_c and $-f_c$ to $(-f_c + f_m)$.
- For positive frequencies, the highest frequency component of the AM wave equals to $(f_c + f_m)$ and the lowest frequency component equals to $(f_c - f_m)$. The difference between these two frequencies defines the **transmission bandwidth B_T** for an AM wave, which is exactly twice the message bandwidth f_m , i.e.

$$B_T = 2f_m$$

[Eq.5.3.21]



Frequency-domain Analysis of AM signal contd.

□ Frequency-domain Analysis of AM signal: (For a single-tone modulation)

- Re-writing the generic mathematical expression of an AM signal $[Eq^n(1)]$ as,

$$s_{AM}(t) = A_c[1 + k_a m(t)] \cos(2\pi f_c t)$$

- For a single-tone modulation $[m(t) = A_m \cos(2\pi f_m t)]$, we get,

$$\begin{aligned}
 s_{AM}(t) &= A_c[1 + k_a \cdot A_m \cos(2\pi f_m t)] \cos(2\pi f_c t) \\
 &= A_c[1 + \mu \cdot \cos(2\pi f_m t)] \cos(2\pi f_c t) \\
 &= A_c \cos(2\pi f_c t) + \mu A_c \cos(2\pi f_m t) \cos(2\pi f_c t) \\
 &= \underbrace{A_c \cos(2\pi f_c t)}_{\text{Carrier}} + \underbrace{\frac{\mu A_c}{2} \cos 2\pi(f_c + f_m)t}_{\text{USB}} + \underbrace{\frac{\mu A_c}{2} \cos 2\pi(f_c - f_m)t}_{\text{LSB}}
 \end{aligned}$$

$f_c + f_m - f_c + f_m$
 $B.W. = 2 f_m$



Frequency-domain Analysis of AM signal contd.

- Taking the Fourier transform of the above Eq^n , we get,

$$\begin{aligned} S_{AM}(f) &= \mathcal{F}[s_{AM}(t)] \\ &= \mathcal{F}\left[A_c \cos(2\pi f_c t) + \frac{\mu A_c}{2} \cos 2\pi(f_c + f_m)t + \frac{\mu A_c}{2} \cos 2\pi(f_c - f_m)t\right] \\ &= \frac{A_c}{2} [\delta(f - f_c) + \delta(f + f_c)] \\ &\quad + \frac{\mu A_c}{4} \{\delta[f - (f_c + f_m)] + \delta[f + (f_c + f_m)]\} \\ &\quad + \frac{\mu A_c}{4} \{\delta[f - (f_c - f_m)] + \delta[f + (f_c - f_m)]\} \quad [\text{Eq.5.3.22}] \end{aligned}$$

Where we have used the relation:

$$\mathcal{F}[\cos(2\pi f_c t)] = \frac{1}{2} [\delta(f - f_c) + \delta(f + f_c)]$$



Frequency-domain Analysis of AM signal contd.

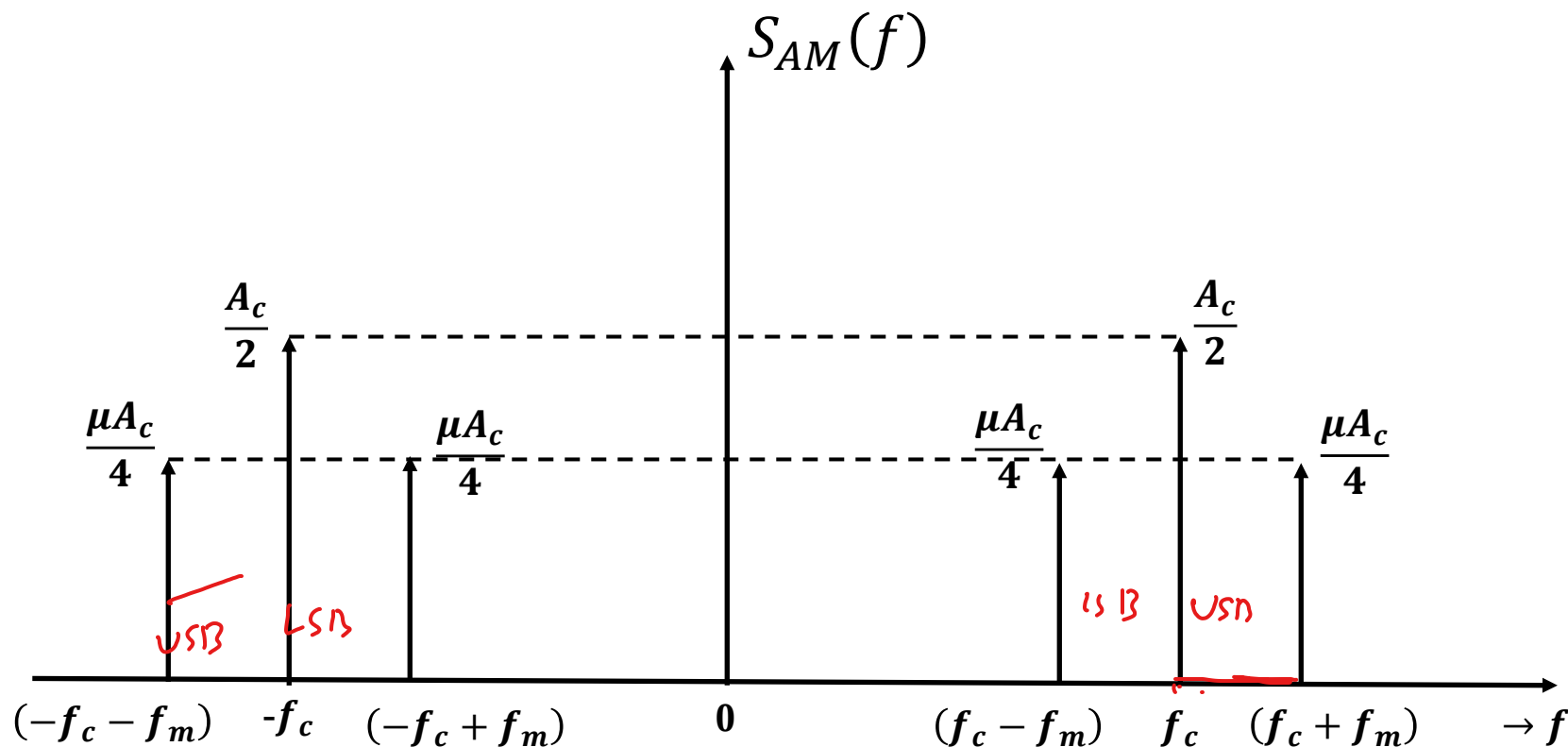


Fig.5.3.7 [Spectrum of AM wave $s_{AM}(t)$]



Power Distribution Analysis of AM signal

□ Power Distribution Analysis of AM signal:

- Re-writing the generic mathematical expression of an AM signal [Eqⁿ(1)] as,

$$\begin{aligned} s_{AM}(t) &= A_c [1 + k_a m(t)] \cos(2\pi f_c t) \\ &= A_c \cos(2\pi f_c t) + k_a A_c m(t) \cos(2\pi f_c t) \end{aligned}$$

- The power distribution in AM signal (**voltage relation**) can be written as;

$$\text{Total Power} = (\text{Unmodulated Carrier Power} + \text{Power of Double Sideband})$$

$$\Rightarrow P_T = P_c + P_{DSB}$$

$$= \left(\frac{A_c}{\sqrt{2}} \right)^2 + \left(\frac{k_a A_c m(t)}{\sqrt{2}} \right)^2$$

$$= \frac{A_c^2}{2} [1 + k_a^2 \overline{m^2(t)}]$$

[Eq.5.3.23]



Power Distribution Analysis of AM signal contd.

- For a single-tone modulation $[m(t) = A_m \cos(2\pi f_m t)]$,

The mean-square value of $m(t) = \overline{m^2(t)} = \frac{A_m^2}{2}$

$$\begin{aligned} \therefore P_T &= \frac{A_c^2}{2} \left[1 + k_a^2 \cdot \frac{A_m^2}{2} \right] = \frac{A_c^2}{2} \left[1 + \frac{\mu^2}{2} \right] \\ &= P_c \left[1 + \frac{\mu^2}{2} \right] \end{aligned} \quad [\text{Eq.5.3.24}]$$

$$\begin{aligned} \Rightarrow P_T &= \frac{A_c^2}{2} + \frac{A_c^2 \mu^2}{4} \\ &= \frac{A_c^2}{2} + \frac{A_c^2 \mu^2}{8} + \frac{A_c^2 \mu^2}{8} \\ &= \underline{P_c} + \underline{P_{LSB}} + \underline{P_{USB}} \quad \text{--- (1)} \end{aligned} \quad [\text{Eq.5.3.25}]$$



Power Distribution Analysis of AM signal contd.

- The power distribution in AM signal (**current relation**) can be written as;

$$\therefore P_T = P_c \left[1 + \frac{\mu^2}{2} \right]$$

2

$$\Rightarrow I_{T,rms}^2 \cdot R = I_{c,rms}^2 \cdot R \left[1 + \frac{\mu^2}{2} \right]$$

$$\Rightarrow I_{T,rms} = I_{c,rms} \sqrt{1 + \frac{\mu^2}{2}}$$

[Eq.5.3.26]

$$\text{Where, Power} = \frac{V_{rms}^2}{R} = I_{rms}^2 \cdot R$$



Maximum Power Efficiency

□ Maximum Power Efficiency/

Transmission Efficiency of AM modulator:

- The power efficiency of an AM modulator is given by;

$$\eta = \frac{\text{Total power in the information bearing sidebands} \checkmark}{\text{Total power in the modulated signal} \checkmark}$$

$$= \frac{P_c \mu^2 / 2}{P_c \left[1 + \frac{\mu^2}{2} \right]} = \frac{\mu^2}{2 + \mu^2} \quad [\text{Eq.5.3.27}]$$



AM For Complex Modulating Signal

□ Analysis of AM With Complex Modulating Signal:

- Practically, the modulating signal is not a single-frequency signal but is a complex/multi-tone or arbitrary waveform, consisting of two or more sinusoidal waves of different frequencies with different amplitudes.

- Let a modulating signal contains two frequencies f_{m1} and f_{m2} i.e.

$$m(t) = A_{m1} \cos(2\pi f_{m1}t) + A_{m2} \cos(2\pi \underline{f_{m2}t}) \quad \checkmark$$

- This signal modulates a single frequency carrier signal,

$$c(t) = A_c \cos(2\pi f_c t) \quad \checkmark$$

- As per the basic definition of amplitude modulation, the AM signal is expressed as;

$$\underline{s_{AM}(t)} = A_c [1 + k_a m(t)] \cos(2\pi f_c t) \quad \checkmark$$



AM For Complex Modulating Signal contd.

$$\begin{aligned}\Rightarrow s_{AM}(t) &= A_c [1 + k_a m(t)] \cos(2\pi f_c t) \\ &= A_c [1 + k_a (A_{m1} \cos(2\pi f_{m1} t) + A_{m2} \cos(2\pi f_{m2} t))] \cos(2\pi f_c t)\end{aligned}$$

[Eq.5.3.28]

- Expanding various terms, we get;

$$\begin{aligned}s_{AM}(t) &= A_c \cos(2\pi f_c t) \\ &\quad + \underline{A_c k_a A_{m1} \cos(2\pi f_{m1} t)} \underline{\cos(2\pi f_c t)} \\ &\quad + \underline{A_c k_a A_{m2} \cos(2\pi f_{m2} t)} \underline{\cos(2\pi f_c t)} \\ &= A_c \cos(2\pi f_c t) \\ &\quad + A_c \mu_1 \cos(2\pi f_{m1} t) \cos(2\pi f_c t) \\ &\quad + A_c \mu_2 \cos(2\pi f_{m2} t) \cos(2\pi f_c t)\end{aligned}$$

[Eq.5.3.29]



AM For Complex Modulating Signal contd.

- Taking the Fourier transform of the above Eqⁿ (2), we get,

$$\begin{aligned}
 S_{AM}(f) &= \mathcal{F}[s_{AM}(t)] \\
 &= \frac{A_c}{2} [\delta(f - f_c) + \delta(f + f_c)] \\
 &\quad + \frac{\mu A_c}{4} \{\delta[f - (f_c + f_{m1})] + \delta[f + (f_c + f_{m1})]\} \quad \checkmark \\
 &\quad + \frac{\mu A_c}{4} \{\delta[f - (f_c - f_{m1})] + \delta[f + (f_c - f_{m1})]\} \quad \checkmark \\
 &\quad + \frac{\mu A_c}{4} \{\delta[f - (f_c + f_{m2})] + \delta[f + (f_c + f_{m2})]\} \quad \checkmark \\
 &\quad + \frac{\mu A_c}{4} \{\delta[f - (f_c - f_{m2})] + \delta[f + (f_c - f_{m2})]\} \quad \checkmark
 \end{aligned}$$

[Eq.5.3.30]

Note: This discussion can be extended to the complex modulating signal comprising of more than two modulating frequencies.



AM For Complex Modulating Signal contd.

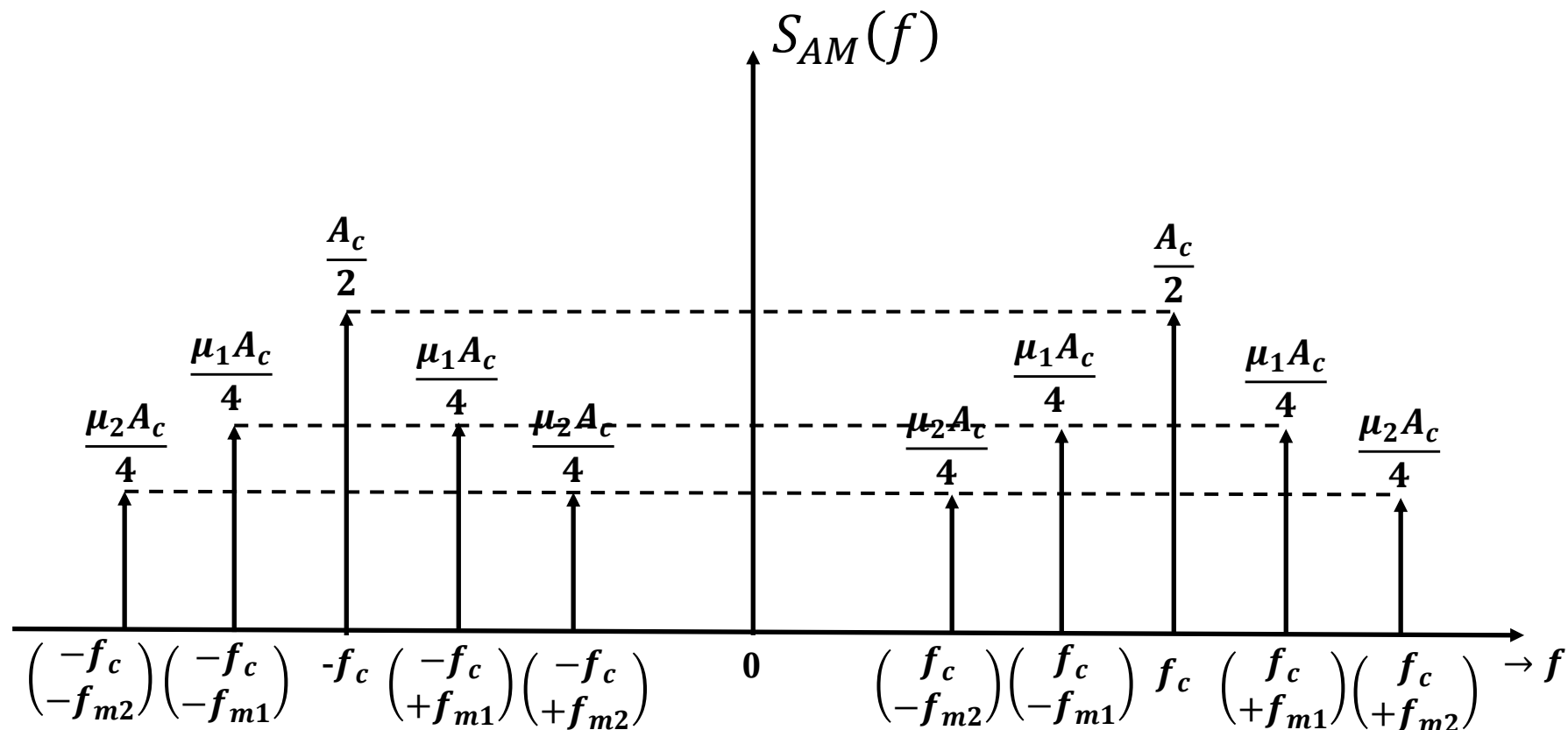


Fig.5.3.8 [Spectrum of AM With Complex Modulating Signal $s_{AM}(t)$]



AM For Complex Modulating Signal contd.

□ Modulation Index For Complex Modulating Signal:

- The Modulation Index for complex AM signal is the square root of the quadratic sum of the individual modulation indices due to the individual modulating signals.

$$\mu_T = \sqrt{\mu_1^2 + \mu_2^2 + \mu_3^2 + \dots} \quad \text{---)} \quad [\text{Eq.5.3.31}]$$

Proof:

- Let $A_{m1}, A_{m2}, A_{m3} \dots$ etc. are the modulating voltages of several individual signals.
- Let A_{mT} be the total modulating voltage due to all modulating signals.

$$\underline{A_{mT}} = \sqrt{\left[\frac{A_{m1}^2}{2} + \frac{A_{m2}^2}{2} + \frac{A_{m3}^2}{2} + \dots \right]} \quad \text{✓}$$

$$\mu = \frac{A_m}{A_c}$$



AM For Complex Modulating Signal contd.

∴ Total Modulation Index

$$\mu_T = \frac{A_{mT}}{\sqrt{A_c^2/2}}$$

$$= \frac{\sqrt{[A_{m1}^2/2 + A_{m2}^2/2 + A_{m3}^2/2 + \dots]}}{\sqrt{A_c^2/2}}$$

$$= \sqrt{[A_{m1}^2/A_c^2 + A_{m2}^2/A_c^2 + A_{m3}^2/A_c^2 + \dots]}$$

$$\mu_T = \sqrt{\mu_1^2 + \mu_2^2 + \mu_3^2 + \dots}$$



AM For Complex Modulating Signal contd.

□ Limitations of AM:

- **Low Transmitted Power Efficiency:** The useful power of the Double sideband is quite small as compared to the total transmitted power of the AM signal. So the useful transmitted power efficiency is quite low.
- **Poor Reception Quality:** AM broadcasting radio stations are assigned transmission bandwidth of 10 KHz only so as to minimize the interference from the adjacent AM broadcasting radio stations, thereby limiting the reception quality.
- **Noisy Signal Reception:** An AM receiver can not distinguish between amplitude variations that contain the desired modulating signal and that represent external noise. Therefore the signal reception is generally noisy.



AM For Complex Modulating Signal contd.

- **Limited Operating Radio Range:** Due to the low transmitted power efficiency, signal can not be transmitted over a long distances without increasing the transmitted power substantially. However, long distance broadcast communication is possible only for short-wave AM because of wave propagation characteristics.

□ Applications of AM:

- AM is widely used in medium and high-frequency bands for long distance radio broadcast applications and aircraft communications in the VHF frequency range.
- AM is also used for two-way mobile radio communication such as Citizen's Band (CB) radio.

